

deliver performance gains of ten times or more over traditional drivers without state-caching technology.

To deal with fan-out at functional test, many manufacturers place several independent but identical testers in parallel. However, this method uses capital equipment very inefficiently, because even under-used instruments are duplicated for each tester. A far more efficient method is to test several devices in parallel on a single, parallel tester with multiple test sockets. A parallel test strategy offers the most efficient use of equipment because more expensive, under-used instruments can be shared among sockets while more common instrumentation can be duplicated per socket for maximum throughput.

Increasing the engineering productivity. Getting a product into volume production and on the market as soon as possible can make the difference between a successful product and a failure. In highly competitive industries, for example, a six-month delay in the shipment of a product can result in more than a 30 per cent decrease in cumulative profits over the entire life-cycle of the product. For such applications, developing a manufacturing test system rapidly is essential, as is quickly adapting a tester after product revisions.

Technologies based on widely available commercial technologies have inherent productivity advantages. PXI, for example, takes advantage of standard Intel and Microsoft platforms to deliver plug-and-play device installation and easy, consistent device configuration. PXI measurement devices also offer increased flexibility over proprietary platforms, because the PXI device makes only the raw measurement and uses software for analysis and decision making. Thus when new products are brought to production, systems using PXI can be quickly changed to add new measurement routines in software, often without requiring hardware changes.

Another way to maximise productivity is to use commercially available software whenever possible instead of

creating custom software packages in-house. Many organisations, for example, write their own test codes instead of using an off-the-shelf product. Spending valuable engineering time recreating software that already exists, reduces the overall productivity and ties up even more resources over time for software maintenance. Today's successful manufacturing organisations concentrate on core competencies such as the design of products and test strategies to achieve efficient use of available resources.

Lowering the overall cost of ownership. The cost of a test system is measured by two variables:

1. Initial cost of the equipment, plus development and/or integration services
2. Cost over time for maintenance and support of the test system

PXI measurement devices use commercially available technologies such as standard integrated circuits for analogue-to-digital conversion and PCI bus interfacing. Because these devices are widely used in other high-volume technologies, such as telecommunications and consumer electronics products, their unit cost is substantially lower than highly customised components. This feature results in measurement functionality at a cost much lower than that of devices in proprietary platforms. In addition, the PXI platform uses components already present in your computer—the processor, memory and monitor—to perform the measurement analysis and display instead of duplicating this functionality in a customised box instrument.

PXI also lowers costs by delivering a platform capable of diverse measurement and control capability. In many systems, the platform for analogue measurements is completely separate from functionality, such as in optical inspection, motion control or continuous data acquisition. PXI, however, integrates this diverse set of measurements into a single hardware platform to lower overall equipment costs and to permit a greater degree of integration between these systems.

Much of the total cost of a test system comes after the initial capital expenditure, because software training and maintenance costs can add significantly to the cost of test. Widely used commercial software have standard training, technical support and a large user base. More the commercial technologies you use in your test system, better the advantage of these cost savings.

Maintenance is often a hidden cost in software development. All software require some amount of maintenance over time. Keeping up with the latest operating systems, for example, is a requirement for compatibility with hardware and other systems in the enterprise. Porting customised software to a new operating system can use up valuable resources and time. Some companies spend several million dollars, for example, to port in-house test software from disk operating system to Windows NT. Using commercial software tools offloads much of the maintenance burden to the software vendor and lowers your overall cost.

To sum up

A manufacturing test system using widely used hardware and software platforms built on commercial technologies and open industry standards can dramatically lower the cost of test. Such a platform increases the test throughput, with fast measurement hardware and software capable of managing multiple test routines in parallel. Engineering productivity will increase, because engineers would take full advantage of powerful software programs to develop tests rapidly. The time required to redesign test systems for new products will decrease with the use of flexible, modular software and hardware platforms. Finally, the cost of the test system, including both the initial capital expenditure and the overall cost of ownership, will be drastically lower, thanks to the leverage of commercial technologies and widely used standards. ●

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